Paper Summary

A Joint Cross-Attention Model for Audio-Visual Fusion in Dimensional Emotion Recognition

English Summary with Figures and Tables

Core Topic

Audio-visual fusion for dimensional emotion recognition using a joint cross-attention model that captures both intra-modal and inter-modal relationships.

Item	Details
Paper Type	Audio-Visual Fusion / Emotion Recognition
Main Modalities	Audio and Visual / facial expressions
Task	Dimensional Emotion Recognition
Output	Continuous Valence and Arousal values
Main Technique	Joint Cross-Attention based on correlations between joint A-V representation and individual modalities
Why It Matters for Our Project	It is directly aligned with the prototype direction: combining speech/audio with facial expressions.

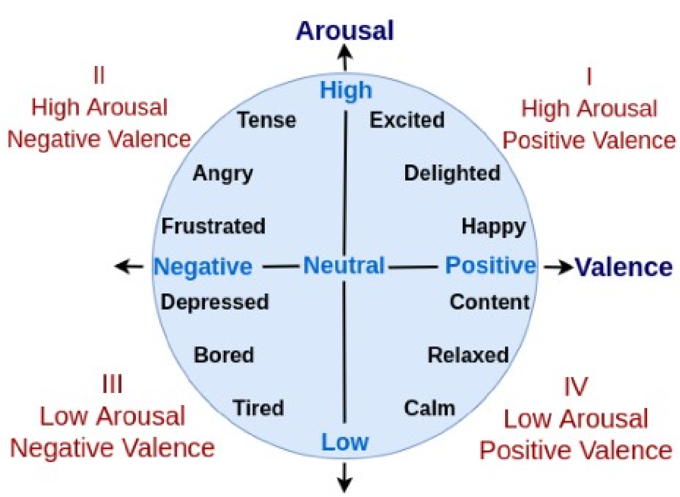

Figure 1. **The valence-arousal space.**

Figure 1. Valence-arousal space used to represent continuous emotional states.

1. Paper Overview

This paper proposes a joint cross-attention model for audio-visual fusion in dimensional emotion recognition. The model uses facial and vocal modalities extracted from videos to predict continuous values of valence and arousal.

The key idea is that audio and visual signals are complementary. When facial information is unclear due to blur, pose variation, or low illumination, audio can provide useful emotional cues. Similarly, when audio is silent or noisy, facial expressions can support the prediction.

Main Contribution

The paper introduces a joint cross-attention fusion mechanism that uses a combined audio-visual representation to attend to both audio and visual features, allowing the model to capture complementary relationships more effectively than simple concatenation or vanilla cross-attention.

2. Problem Statement

Many previous audio-visual emotion recognition systems combine audio and visual features using simple feature concatenation, recurrent networks, or conventional attention mechanisms. However, these methods do not fully capture the inter-modal relationships between facial and vocal cues.

- Feature concatenation may ignore the complementary relationship between audio and visual modalities.
- Recurrent models can capture temporal patterns but may not explicitly model cross-modal interactions.
- Vanilla cross-attention considers relations between modalities, but it may not sufficiently use the joint audio-visual representation.
- Real-world emotion recognition is difficult because videos often contain noise, pose changes, illumination variation, and ambiguous emotional annotations.

3. Proposed Solution

The proposed solution is a Joint Cross-Attention model for audio-visual fusion. Instead of directly concatenating audio and visual features, the model first extracts separate audio and visual feature representations, then builds a joint audio-visual representation.

This joint representation is used inside the cross-attention module to compute attention between the combined A-V features and the individual audio and visual features. As a result, each modality can attend to both itself and the other modality.

Component	Role in the Model
Visual stream	Detects, tracks, and aligns faces, then extracts visual features using a visual backbone.
Audio stream	Extracts vocal information using resampling, MFCC/spectrogram processing, and an audio backbone.
Joint representation	Combines audio and visual features into one A-V representation.

Joint cross-attention	Models intra-modal and inter-modal relationships using correlations between joint and individual features.
Prediction head	Predicts continuous valence and arousal values.

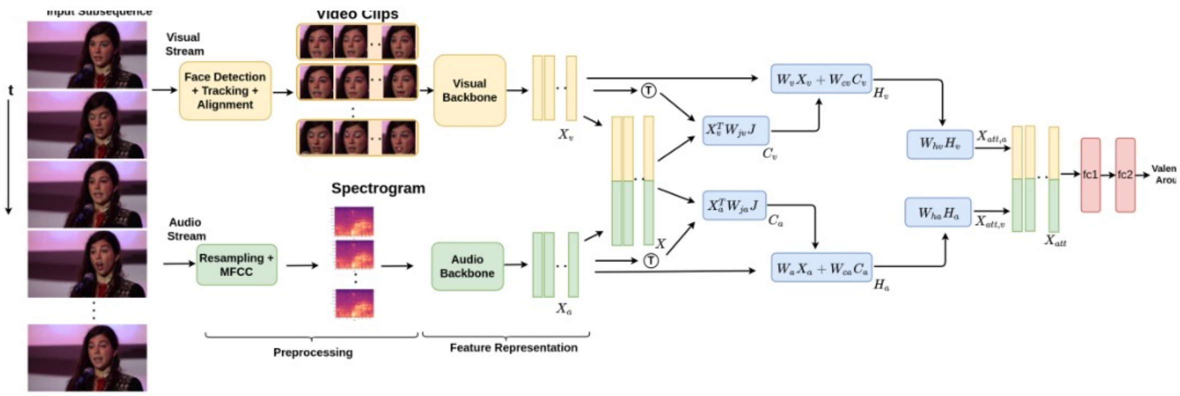

Figure 2. **Joint cross-attention model proposed for A-V fusion (training mode).**

Figure 2. Proposed joint cross-attention architecture for audio-visual fusion.

4. Understanding Valence and Arousal

The paper uses dimensional emotion recognition instead of simple categorical emotion labels. In this setting, emotions are represented using two continuous dimensions: valence and arousal.

Dimension	Meaning	Examples
Valence	Measures how positive or negative an emotional state is.	Negative: sad, angry; Positive: happy, content
Arousal	Measures the intensity or activation level of the emotion.	High: excited, tense; Low: calm, tired

This representation is more flexible than a fixed set of emotion classes because it can model a wide range of emotional states on a continuous scale.

5. Architecture Details

5.1 Visual Network

The visual stream focuses on facial expressions. Faces are detected, tracked, aligned, and passed through a visual backbone such as I3D, R3D, ResNet18 + GRU, or I3D + TCN. The visual backbone extracts spatial and temporal facial features.

5.2 Audio Network

The audio stream extracts vocal features from the video. The signal is converted into spectrogram-based features, and ResNet18 is used as the audio backbone. These audio features capture vocal cues related to the emotional state.

5.3 Joint Cross-Attention Module

Audio features and visual features are concatenated to form a joint A-V representation. The model then computes correlations between this joint representation and each individual modality. This enables the system to identify which audio and visual features are most relevant for predicting valence and arousal.

Why Joint Cross-Attention Is Important

The model does not only allow audio to interact with video. It also allows each modality to attend to itself through the joint representation, which helps capture both intra-modal and inter-modal relationships.

6. Dataset and Evaluation Metric

The experiments are conducted on the AffWild2 dataset, one of the largest in-the-wild affective computing datasets. It contains videos with continuous valence and arousal annotations collected from real-world YouTube videos.

- The paper focuses on valence-arousal estimation.
- Annotations are continuous values in the range [-1, 1].

- The dataset contains challenging real-world conditions such as illumination changes, pose variation, and background noise.
- The evaluation metric is Concordance Correlation Coefficient (CCC), which is commonly used for dimensional emotion recognition.

Metric

CCC measures the agreement between predicted values and ground-truth annotations. Higher CCC means better prediction consistency with the true valence/arousal labels.

7. Experimental Results

The paper evaluates the proposed model through ablation studies and comparison with state-of-the-art methods. The results show that joint cross-attention improves fusion performance, especially for valence prediction.

7.1 Ablation Study

Table 1 compares different fusion strategies and visual backbones on the AffWild2 development set. Joint Cross-Attention generally outperforms simple feature concatenation and vanilla cross-attention. The best valence score is obtained using I3D + TCN with Joint Cross-Attention.

Table 1. **Performance of our approach model with various components on the Affwild2 dataset. Resnet18 [7] is used to extract A features in all experiments.**

V Backbone	Fusion	Valence	Arousal
I3D	Feature Concatenation	0.531	0.468
R3D	Feature Concatenation	0.517	0.493
I3D	Cross-Attention [36]	0.541	0.517
I3D	Leader-Follower [40]	0.592	0.521
Resnet18 + GRU	Joint Cross Attention (Ours)	0.632	0.520
R3D	Joint Cross-Attention (Ours)	0.642	**0.592**
I3D	Joint Cross-Attention (Ours)	0.657	0.580
I3D + TCN	Joint Cross-Attention (Ours)	**0.663**	0.584

Table 1. Ablation study showing the effect of different backbones and fusion methods.

7.2 Validation Set Comparison

Table 2 compares the proposed fusion method with other audio-visual fusion methods on the AffWild2 development set. The proposed method achieves the best valence fusion score among the listed methods.

Table 2. **CCC performance of the proposed and state-of-art methods for A-V fusion on the Affwild2 development set.**

Method – A/V backbone	Valence			Arousal		
	Audio	Visual	Fusion	Audio	Visual	Fusion
Kuhnke [21], FGW 2020 – A: Resnet18; V: R(2plus1)D	0.351	0.449	0.493	0.356	0.565	0.604
Zhang [49], ICCVW 2021 – A: VGGish; V: Resnet50 + TCN	-	0.405	0.457	-	0.635	**0.645**
Rajasekhar [36], FG 2021 – A: Resnet18; V: I3D + TCN	0.351	0.417	0.552	0.356	0.539	0.531
Joint Cross-Attention (Ours) – A: Resnet18; V: I3D + TCN	0.351	0.417	**0.663**	0.356	0.539	0.584

Table 2. CCC performance comparison on the AffWild2 development set.

7.3 Test Set Comparison

Table 3 reports the test set comparison. The proposed model significantly improves over the baseline. The baseline achieves 0.180 for valence and 0.170 for arousal, while the proposed model achieves 0.374 for valence and 0.363 for arousal.

Table 3. **CCC performance of the proposed and state-of-art methods for A-V fusion on Affwild2 test set.**

Method	**Modalities Used**	**Valence**	**Arousal**	**Mean**
Situ-RUCAIM3 [28]	Audio, Visual	0.606	0.596	0.601
FlyingPigs [48]	Audio, Visual, Text	0.520	0.602	0.561
PRL [32]	Visual	0.450	0.445	0.448
HSE-NN [38]	Visual	0.417	0.454	0.436
AU-NO [10]	Audio, Visual	0.418	0.407	0.413
Joint Cross-Attention (Ours)	Audio, Visual	**0.374**	**0.363**	**0.369**
Baseline	Visual	0.180	0.170	0.175

Table 3. Test set comparison with state-of-the-art methods.

Metric	Baseline	Joint Cross-Attention	Improvement
Valence	0.180	0.374	More than 2x higher
Arousal	0.170	0.363	More than 2x higher
Mean	0.175	0.369	Strong overall improvement

8. Key Findings

- Joint Cross-Attention improves audio-visual fusion compared with simple feature concatenation.
- Using a joint A-V representation helps capture complementary relationships between audio and facial modalities.
- The model is particularly strong for valence prediction.
- The proposed method improves significantly over the baseline on the AffWild2 test set.
- The paper is directly relevant to prototypes that combine speech/audio and facial expressions.

9. Research Gap

Although the paper provides a strong fusion strategy for audio-visual emotion recognition, it does not fully address several practical challenges that appear in real-world multimodal systems.

- The model assumes that audio and visual modalities are available during inference.
- It does not explicitly handle missing modality scenarios.
- It does not include a modality quality assessment mechanism to decide when audio or visual information is unreliable.
- It does not directly solve modality imbalance, where one modality dominates the prediction.
- It focuses on audio and visual modalities only and does not include language, physiological signals, motion, touch, or environmental sensors.

Research Gap Statement

Although Joint Cross-Attention effectively models audio-visual interactions for dimensional emotion recognition, it assumes that both modalities are available and reliable. Integrating cross-attention fusion with missing-modality robustness and modality-aware balancing remains an important practical research gap.

10. Relevance to Our Multimodal CIR Project

This paper is one of the most relevant references for our prototype because our prototype direction focuses on combining audio with facial expressions. The paper provides a strong theoretical and architectural basis for designing an audio-visual fusion module.

Project Aspect	How This Paper Helps
Prototype modalities	Supports the audio + facial expression direction.
Fusion strategy	Suggests using cross-attention rather than simple concatenation.
Emotion understanding	Uses valence and arousal to represent emotional states continuously.
Research gap	Motivates adding missing-modality handling and modality balance mechanisms.

11. Final Conclusion

The paper presents an effective joint cross-attention model for audio-visual fusion in dimensional emotion recognition. By using a joint audio-visual representation, the model captures both intra-modal and inter-modal relationships and improves performance over the baseline. For our project, this paper provides a strong foundation for the audio-face prototype, while also exposing a clear gap: the need for reliable fusion under noisy, imbalanced, or missing modalities.